FTP AND SNMP NETWORK MANAGEMENT ASSIGNMENT

1. Install FileZilla Server on a Windows machine, create a user account, and set up a shared folder for FTP access.

Setting	Values
FTP server address	192.168.1.50
FTP port	21
Username	ftpstudent
Password	Ftp@2026!
Shared folder	C:\FTP-Shared
Permission	Read, Write, and List

Setup steps

1. Install FileZilla Server on Windows and open its administration interface.
2. Create a user account named ftpstudent and assign a strong password.
3. Add the folder C:\FTP-Shared as the user's shared directory.
4. Grant the user permission to list files and read or write files in the shared folder.
5. Allow the required FTP connections through Windows Firewall for the local test network.

2. Use FileZilla Client to connect to the FTP server with the created credentials and upload an image file from the computer to the shared folder.

Connection details

Field	Values
Host	192.168.1.50
Username	ftpstudent
Password	Ftp@2026!
Port	21
Protocol	FTP

Uploaded file: network_diagram.jpg

Remote destination: /FTP-Shared/network_diagram.jpg

3. Download the uploaded file from the FTP server to a different folder and verify that its file size matches the original.

File detail	Sample value
Original file	network_diagram.jpg
Original location	C:\Users\Student\Pictures\network_diagram.jpg
Original file size	245 KB
Downloaded location	C:\Users\Student\Downloads\FTP-Test\network_diagram.jpg
Downloaded file size	245 KB
Verification	File sizes match

Explanation: The file is downloaded from the FTP shared folder to a different local folder. The original and downloaded file sizes are compared. Since both sample sizes are 245 KB, the transfer appears complete. A checksum comparison, such as SHA-256, would provide stronger verification that the file contents are identical.

4. List 3 real-world scenarios where SNMP is used in network management for apps or services such as ISPs, campus Wi-Fi, or large websites.

Real-world scenario	How SNMP is used
1. ISP network monitoring	An Internet Service Provider monitors routers, switches, and network links for bandwidth usage, interface errors, and device uptime.
2. College or campus Wi-Fi	Network administrators monitor wireless access points and switches to check device status, traffic levels, and possible failures across campus buildings.
3. Data centre hosting websites or apps	Administrators monitor servers, network switches, routers, and power-related equipment to detect high resource usage, interface problems, or downtime.

Explanation: SNMP (Simple Network Management Protocol) allows network management systems to collect status and performance information from supported devices. This helps administrators detect faults and monitor network health.

5. Create a table comparing FTP and HTTP across at least five points: port number, security, data transfer method, use cases, and authentication.

Comparison point	FTP	HTTP
Default port	Port 21 for control; traditional active/passive data connections use separate data ports.	Port 80 by default.
Security	Traditional FTP is unencrypted; FTPS or SFTP should be used for secure file transfers.	HTTP is unencrypted; HTTPS uses TLS encryption.
Data transfer method	Designed to upload, download, and manage files between a client and server.	Transfers web resources such as HTML pages, images, scripts, and API data.
Common use cases	Transferring files to and from a file server or website hosting server.	Loading websites and communicating with web applications and APIs.
Authentication	Can use a username and password; some servers also allow anonymous access.	Usually uses application-based login, cookies, tokens, or other web authentication methods.

Conclusion: FTP is mainly designed for file transfer, while HTTP is designed for delivering web content and supporting web applications. For secure communication, use FTPS or SFTP instead of plain FTP, and HTTPS instead of plain HTTP.